

THREE BRILLHART–LEHMER–SELFIDGE PRIMALITY PROOFS FOR WAGSTAFF NUMBERS

ALEXEY DOLOTOV

ABSTRACT. The Wagstaff numbers $W_p = (2^p + 1)/3$ for odd primes p are the natural +1 companions of the Mersenne numbers. Published primality proofs for W_p have relied on general-purpose elliptic-curve primality proving (ECPP); Chebyshev/Lucas-type tests, while available as compositeness criteria, have remained conjectural on the sufficiency side. We give fully verified primality proofs of W_{2617} (788 digits), W_{10501} (3161 digits), and W_{12391} (3730 digits), independent of ECPP. The proofs apply the Brillhart–Lehmer–Selfridge (BLS) $N - 1$ framework [3] to the cyclotomic decomposition $2^{p-1} - 1 = \prod_{d|p-1} \Phi_d(2)$, with factorisations drawn from the Cunningham project tables [4] and direct ECM/ $\rho/p-1$ computation. As an independent sanity check, the Chua $N+1$ congruence $\omega_3^{(W_p+1)/2} \equiv -1 \pmod{W_p}$ — the $a = 3$ case of the Chua framework [5] with $\omega_3 = 3 + 2\sqrt{2} \in \mathbb{Z}[\sqrt{2}]$, a necessary condition for Wagstaff primality — is verified at each W_p .

The method requires $p - 1$ to be sufficiently smooth that enough cyclotomic factors $\Phi_d(2)$ have known factorisations; in the current Cunningham tables [4] the exponents $p = 10501$ and $p = 12391$ are the only Wagstaff prime candidates above 2617 meeting this ceiling. Every step is reproducible from the scripts archived with this paper; all cofactor primalities are certified unconditionally by APR-CL [1, 6]. The method does not depend on the Chebyshev sufficiency conjecture; that conjecture is the subject of a companion paper [9].

1. INTRODUCTION

1.1. Wagstaff numbers and the ECPP era. For an odd prime p , the *Wagstaff number* is

$$W_p = \frac{2^p + 1}{3}.$$

The sequence (W_p) is indexed in OEIS as A000979. All W_p with $p \leq 42737$ have been proved prime or composite; the exponents $p \in \{5, 7, 11, 13, 17, 19, 23, 31, 43, 61, 79, 101, 127, 167, 191, 239, 263, 283, 347, 359, 383, 431, 461, 487, 509, 521, 541, 563, 593, 641, 673, 707, 743, 769, 811, 821, 883, 937, 971, 1019, 1031, 1073, 1103, 1151, 1163, 1223, 1271, 1283, 1327, 1367, 1397, 1423, 1483, 1511, 1547, 1571, 1619, 1691, 1709, 1747, 1783, 1811, 1867, 1907, 1931, 1979, 1997, 2011, 2039, 2063, 2083, 2111, 2147, 2171, 2207, 2231, 2279, 2291, 2339, 2381, 2411, 2447, 2471, 2507, 2531, 2579, 2617, 2651, 2671, 2711, 2747, 2771, 2819, 2831, 2879, 2911, 2951, 2981, 3011, 3047, 3071, 3119, 3147, 3171, 3211, 3247, 3271, 3319, 3347, 3371, 3419, 3447, 3471, 3519, 3547, 3571, 3619, 3647, 3671, 3719, 3747, 3771, 3819, 3847, 3871, 3919, 3947, 3971, 4019, 4047, 4071, 4119, 4147, 4171, 4219, 4247, 4271, 4319, 4347, 4371, 4419, 4447, 4471, 4519, 4547, 4571, 4619, 4647, 4671, 4719, 4747, 4771, 4819, 4847, 4871, 4919, 4947, 4971, 5019, 5047, 5071, 5119, 5147, 5171, 5219, 5247, 5271, 5319, 5347, 5371, 5419, 5447, 5471, 5519, 5547, 5571, 5619, 5647, 5671, 5719, 5747, 5771, 5819, 5847, 5871, 5919, 5947, 5971, 6019, 6047, 6071, 6119, 6147, 6171, 6219, 6247, 6271, 6319, 6347, 6371, 6419, 6447, 6471, 6519, 6547, 6571, 6619, 6647, 6671, 6719, 6747, 6771, 6819, 6847, 6871, 6919, 6947, 6971, 7019, 7047, 7071, 7119, 7147, 7171, 7219, 7247, 7271, 7319, 7347, 7371, 7419, 7447, 7471, 7519, 7547, 7571, 7619, 7647, 7671, 7719, 7747, 7771, 7819, 7847, 7871, 7919, 7947, 7971, 8019, 8047, 8071, 8119, 8147, 8171, 8219, 8247, 8271, 8319, 8347, 8371, 8419, 8447, 8471, 8519, 8547, 8571, 8619, 8647, 8671, 8719, 8747, 8771, 8819, 8847, 8871, 8919, 8947, 8971, 9019, 9047, 9071, 9119, 9147, 9171, 9219, 9247, 9271, 9319, 9347, 9371, 9419, 9447, 9471, 9519, 9547, 9571, 9619, 9647, 9671, 9719, 9747, 9771, 9819, 9847, 9871, 9919, 9947, 9971, 10019, 10047, 10071, 10119, 10147, 10171, 10219, 10247, 10271, 10319, 10347, 10371, 10419, 10447, 10471, 10519, 10547, 10571, 10619, 10647, 10671, 10719, 10747, 10771, 10819, 10847, 10871, 10919, 10947, 10971, 11019, 11047, 11071, 11119, 11147, 11171, 11219, 11247, 11271, 11319, 11347, 11371, 11419, 11447, 11471, 11519, 11547, 11571, 11619, 11647, 11671, 11719, 11747, 11771, 11819, 11847, 11871, 11919, 11947, 11971, 12019, 12047, 12071, 12119, 12147, 12171, 12219, 12247, 12271, 12319, 12347, 12371, 12419, 12447, 12471, 12519, 12547, 12571, 12619, 12647, 12671, 12719, 12747, 12771, 12819, 12847, 12871, 12919, 12947, 12971, 13019, 13047, 13071, 13119, 13147, 13171, 13219, 13247, 13271, 13319, 13347, 13371, 13419, 13447, 13471, 13519, 13547, 13571, 13619, 13647, 13671, 13719, 13747, 13771, 13819, 13847, 13871, 13919, 13947, 13971, 14019, 14047, 14071, 14119, 14147, 14171, 14219, 14247, 14271, 14319, 14347, 14371, 14419, 14447, 14471, 14519, 14547, 14571, 14619, 14647, 14671, 14719, 14747, 14771, 14819, 14847, 14871, 14919, 14947, 14971, 15019, 15047, 15071, 15119, 15147, 15171, 15219, 15247, 15271, 15319, 15347, 15371, 15419, 15447, 15471, 15519, 15547, 15571, 15619, 15647, 15671, 15719, 15747, 15771, 15819, 15847, 15871, 15919, 15947, 15971, 16019, 16047, 16071, 16119, 16147, 16171, 16219, 16247, 16271, 16319, 16347, 16371, 16419, 16447, 16471, 16519, 16547, 16571, 16619, 16647, 16671, 16719, 16747, 16771, 16819, 16847, 16871, 16919, 16947, 16971, 17019, 17047, 17071, 17119, 17147, 17171, 17219, 17247, 17271, 17319, 17347, 17371, 17419, 17447, 17471, 17519, 17547, 17571, 17619, 17647, 17671, 17719, 17747, 17771, 17819, 17847, 17871, 17919, 17947, 17971, 18019, 18047, 18071, 18119, 18147, 18171, 18219, 18247, 18271, 18319, 18347, 18371, 18419, 18447, 18471, 18519, 18547, 18571, 18619, 18647, 18671, 18719, 18747, 18771, 18819, 18847, 18871, 18919, 18947, 18971, 19019, 19047, 19071, 19119, 19147, 19171, 19219, 19247, 19271, 19319, 19347, 19371, 19419, 19447, 19471, 19519, 19547, 19571, 19619, 19647, 19671, 19719, 19747, 19771, 19819, 19847, 19871, 19919, 19947, 19971, 20019, 20047, 20071, 20119, 20147, 20171, 20219, 20247, 20271, 20319, 20347, 20371, 20419, 20447, 20471, 20519, 20547, 20571, 20619, 20647, 20671, 20719, 20747, 20771, 20819, 20847, 20871, 20919, 20947, 20971, 21019, 21047, 21071, 21119, 21147, 21171, 21219, 21247, 21271, 21319, 21347, 21371, 21419, 21447, 21471, 21519, 21547, 21571, 21619, 21647, 21671, 21719, 21747, 21771, 21819, 21847, 21871, 21919, 21947, 21971, 22019, 22047, 22071, 22119, 22147, 22171, 22219, 22247, 22271, 22319, 22347, 22371, 22419, 22447, 22471, 22519, 22547, 22571, 22619, 22647, 22671, 22719, 22747, 22771, 22819, 22847, 22871, 22919, 22947, 22971, 23019, 23047, 23071, 23119, 23147, 23171, 23219, 23247, 23271, 23319, 23347, 23371, 23419, 23447, 23471, 23519, 23547, 23571, 23619, 23647, 23671, 23719, 23747, 23771, 23819, 23847, 23871, 23919, 23947, 23971, 24019, 24047, 24071, 24119, 24147, 24171, 24219, 24247, 24271, 24319, 24347, 24371, 24419, 24447, 24471, 24519, 24547, 24571, 24619, 24647, 24671, 24719, 24747, 24771, 24819, 24847, 24871, 24919, 24947, 24971, 25019, 25047, 25071, 25119, 25147, 25171, 25219, 25247, 25271, 25319, 25347, 25371, 25419, 25447, 25471, 25519, 25547, 25571, 25619, 25647, 25671, 25719, 25747, 25771, 25819, 25847, 25871, 25919, 25947, 25971, 26019, 26047, 26071, 26119, 26147, 26171, 26219, 26247, 26271, 26319, 26347, 26371, 26419, 26447, 26471, 26519, 26547, 26571, 26619, 26647, 26671, 26719, 26747, 26771, 26819, 26847, 26871, 26919, 26947, 26971, 27019, 27047, 27071, 27119, 27147, 27171, 27219, 27247, 27271, 27319, 27347, 27371, 27419, 27447, 27471, 27519, 27547, 27571, 27619, 27647, 27671, 27719, 27747, 27771, 27819, 27847, 27871, 27919, 27947, 27971, 28019, 28047, 28071, 28119, 28147, 28171, 28219, 28247, 28271, 28319, 28347, 28371, 28419, 28447, 28471, 28519, 28547, 28571, 28619, 28647, 28671, 28719, 28747, 28771, 28819, 28847, 28871, 28919, 28947, 28971, 29019, 29047, 29071, 29119, 29147, 29171, 29219, 29247, 29271, 29319, 29347, 29371, 29419, 29447, 29471, 29519, 29547, 29571, 29619, 29647, 29671, 29719, 29747, 29771, 29819, 29847, 29871, 29919, 29947, 29971, 30019, 30047, 30071, 30119, 30147, 30171, 30219, 30247, 30271, 30319, 30347, 30371, 30419, 30447, 30471, 30519, 30547, 30571, 30619, 30647, 30671, 30719, 30747, 30771, 30819, 30847, 30871, 30919, 30947, 30971, 31019, 31047, 31071, 31119, 31147, 31171, 31219, 31247, 31271, 31319, 31347, 31371, 31419, 31447, 31471, 31519, 31547, 31571, 31619, 31647, 31671, 31719, 31747, 31771, 31819, 31847, 31871, 31919, 31947, 31971, 32019, 32047, 32071, 32119, 32147, 32171, 32219, 32247, 32271, 32319, 32347, 32371, 32419, 32447, 32471, 32519, 32547, 32571, 32619, 32647, 32671, 32719, 32747, 32771, 32819, 32847, 32871, 32919, 32947, 32971, 33019, 33047, 33071, 33119, 33147, 33171, 33219, 33247, 33271, 33319, 33347, 33371, 33419, 33447, 33471, 33519, 33547, 33571, 33619, 33647, 33671, 33719, 33747, 33771, 33819, 33847, 33871, 33919, 33947, 33971, 34019, 34047, 34071, 34119, 34147, 34171, 34219, 34247, 34271, 34319, 34347, 34371, 34419, 34447, 34471, 34519, 34547, 34571, 34619, 34647, 34671, 34719, 34747, 34771, 34819, 34847, 34871, 34919, 34947, 34971, 35019, 35047, 35071, 35119, 35147, 35171, 35219, 35247, 35271, 35319, 35347, 35371, 35419, 35447, 35471, 35519, 35547, 35571, 35619, 35647, 35671, 35719, 35747, 35771, 35819, 35847, 35871, 35919, 35947, 35971, 36019, 36047, 36071, 36119, 36147, 36171, 36219, 36247, 36271, 36319, 36347, 36371, 36419, 36447, 36471, 36519, 36547, 36571, 36619, 36647, 36671, 36719, 36747, 36771, 36819, 36847, 36871, 36919, 36947, 36971, 37019, 37047, 37071, 37119, 37147, 37171, 37219, 37247, 37271, 37319, 37347, 37371, 37419, 37447, 37471, 37519, 37547, 37571, 37619, 37647, 37671, 37719, 37747, 37771, 37819, 37847, 37871, 37919, 37947, 37971, 38019, 38047, 38071, 38119, 38147, 38171, 38219, 38247, 38271, 38319, 38347, 38371, 38419, 38447, 38471, 38519, 38547, 38571, 38619, 38647, 38671, 38719, 38747, 38771, 38819, 38847, 38871, 38919, 38947, 38971, 39019, 39047, 39071, 39119, 39147, 39171, 39219, 39247, 39271, 39319, 39347, 39371, 39419, 39447, 39471, 39519, 39547, 39571, 39619, 39647, 39671, 39719, 39747, 39771, 39819, 39847, 39871, 39919, 39947, 39971, 40019, 40047, 40071, 40119, 40147, 40171, 40219, 40247, 40271, 40319, 40347, 40371, 40419, 40447, 40471, 40519, 40547, 40571, 40619, 40647, 40671, 40719, 40747, 40771, 40819, 40847, 40871, 40919, 40947, 40971, 41019, 41047, 41071, 41119, 41147, 41171, 41219, 41247, 41271, 41319, 41347, 41371, 41419, 41447, 41471, 41519, 41547, 41571, 41619, 41647, 41671, 41719, 41747, 41771, 41819, 41847, 41871, 41919, 41947, 41971, 42019, 42047, 42071, 42119, 42147, 42171, 42219, 42247, 42271, 42319, 42347, 42371, 42419, 42447, 42471, 42519, 42547, 42571, 42619, 42647, 42671, 42719, 42747, 42771, 42819, 42847, 42871, 42919, 42947, 42971, 43019, 43047, 43071, 43119, 43147, 43171, 43219, 43247, 43271, 43319, 43347, 43371, 43419, 43447, 43471, 43519, 43547, 43571, 43619, 43647, 43671, 43719, 43747, 43771, 43819, 43847, 43871, 43919, 43947, 43971, 44019, 44047, 44071, 44119, 44147, 44171, 44219, 44247, 44271, 44319, 44347, 44371, 44419, 44447, 44471, 44519, 44547, 44571, 44619, 44647, 44671, 44719, 44747, 44771, 44819, 44847, 44871, 44919, 44947, 44971, 45019, 45047, 45071, 45119, 45147, 45171, 45219, 45247, 45271, 45319, 45347, 45371, 45419, 45447, 45471, 45519, 45547, 45571, 45619, 45647, 45671, 45719, 45747, 45771, 45819, 45847, 45871, 45919, 45947, 45971, 46019, 46047, 46071, 46119, 46147, 46171, 46219, 46247, 46271, 46319, 46347, 46371, 46419, 46447, 46471, 46519, 46547, 46571, 46619, 46647, 46671, 46719, 46747, 46771, 46819, 46847, 46871, 46919, 46947, 46971, 47019, 47047, 47071, 47119, 47147, 47171, 47219, 47247, 47271, 47319, 47347, 47371, 47419, 47447, 47471, 47519, 47547, 47571, 47619, 47647, 47671, 47719, 47747, 47771, 47819, 47847, 47871, 47919, 47947, 47971, 48019, 48047, 48071, 48119, 48147, 48171, 48219, 48247, 48271, 48319, 48347, 48371, 48419, 48447, 48471, 48519, 48547, 48571, 48619, 48647, 48671, 48719, 48747, 48771, 48819, 48847, 48871, 48919, 48947, 48971, 49019, 49047, 49071, 49119, 49147, 49171, 49219, 49247, 49271, 49319, 49347, 49371, 49419, 49447, 49471, 49519, 49547, 49571, 49619, 49647, 49671, 49719, 49747, 49771, 49819, 49847, 49871, 49919, 49947, 49971, 50019, 50047, 50071, 50119, 50147, 50171, 50219, 50247, 50271, 50319, 50347, 50371, 50419, 50447, 50471, 50519, 50547, 50571, 50619, 50647, 50671, 50719, 50747, 50771, 50819, 50847, 50871, 50919, 50947, 50971, 51019, 51047, 51071, 51119, 51147, 51171, 51219, 51247, 51271, 51319, 51347, 51371, 51419, 51447, 51471, 51519, 51547, 51571, 51619, 51647, 51671, 51719, 51747, 51771, 51819, 51847, 51871, 51919, 51947, 51971, 52019, 52047, 52071, 52119, 52147, 52171, 52219, 52247, 52271, 52319, 52347, 52371, 52419, 52447, 52471, 52519, 52547, 52571, 52619, 52647, 52671, 52719, 52747, 52771, 52819, 52847, 52871, 52919, 52947, 52971, 53019, 53047, 53071, 53119, 53147, 53171, 53219, 53247, 53271, 53319, 53347, 53371, 53419, 53447, 53471, 53519, 53547, 53571, 53619, 53647, 53671, 53719, 53747, 53771, 53819, 53847, 53871, 53919, 53947, 53971, 54019, 54047, 54071, 54119, 54147, 54171, 54219, 54247, 54271, 54319, 54347, 54371, 54419, 54447, 54471, 54519, 54547, 54571, 54619, 54647, 54671, 54719, 54747, 54771, 54819, 54847, 54871, 54919, 54947, 54971, 55019, 55047, 55071, 55119, 55147, 55171, 55219, 55247, 55271, 55319, 55347, 55371, 55419, 55447, 55471, 5551$

test of comparable simplicity is known for Wagstaff numbers. A natural Chebyshev/Lucas-type candidate — the single congruence

$$\omega_3^{(W_p+1)/2} \equiv -1 \pmod{W_p}, \quad \omega_3 = 3 + 2\sqrt{2},$$

in $\mathbb{Z}[\sqrt{2}]$ (the $a = 3$ case of the Chua framework [5]) — is known to hold for every Wagstaff prime, but its sufficiency remains conjectural; see the companion paper [9].

In the absence of a deterministic standalone test, the Brillhart–Lehmer–Selfridge (BLS) framework [3] provides unconditional primality proofs whenever $N \pm 1$ has a “large enough” known factored portion. BLS is classical and venerable, but the specific exponents $p = 2617, 10501, 12391$ have not previously been put through BLS in the published literature; the ECPP proofs already in the tables are of a completely different nature, and do not depend on factorisations of $W_p \pm 1$.

1.2. This paper. We give three unconditional BLS primality proofs, one per exponent. The proofs proceed by:

- (1) Establishing that Condition (II), $\omega_3^{(W_p+1)/2} \equiv -1 \pmod{W_p}$, holds for every Wagstaff prime W_p (Proposition 2.4) — this is the $a = 3$ case of the Chua framework [5] and follows from $W_p \equiv 3 \pmod{8}$.
- (2) Recording two derived criteria: a general $N + 1$ criterion via ω_3 (Theorem 3.2), useful when enough of W_{p-2} is factored to give $F_+ > \sqrt{W_p}$; and its small- p specialisation (Corollary 3.4), which collapses to “Condition (II) iff W_p prime” whenever W_{p-2} itself is prime (Remark 3.5).
- (3) For the three target exponents $p \in \{2617, 10501, 12391\}$, applying the BLS $N - 1$ criterion (Theorem 3.6) to the cyclotomic decomposition $2^{p-1} - 1 = \prod_{d|p-1} \Phi_d(2)$, harvesting factored portions from the Cunningham project tables [4] plus direct ECM/ $p-1$ /Pollard- ρ computation. Every cofactor primality is certified by APR-CL [1, 6].
- (4) Verifying Condition (II) independently at each of the three W_p as a redundant primality witness — consistent with Proposition 2.4 and a useful debugging check on the modular-arithmetic implementation.

The resulting proofs are short (Section 4), fully unconditional, and cleanly separated from the Chebyshev sufficiency conjecture (not used, not needed). All computations are reproducible from the scripts described in Section 5.

1.3. Notation. Throughout, $p \geq 5$ denotes an odd prime and $W_p = (2^p + 1)/3$. For an integer or algebraic integer x coprime to a modulus m , $\text{ord}_m(x)$ denotes the multiplicative order of x modulo m ; $v_q(\cdot)$ denotes the q -adic valuation. We work in $\mathbb{Z}[\sqrt{2}]$, with fundamental unit $\alpha = 1 + \sqrt{2}$ and Chebyshev base $\omega_3 = \alpha^2 = 3 + 2\sqrt{2}$. The Pell sequences (U_n) and (V_n) are defined by $\alpha^n = V_n/2 + U_n\sqrt{2}$, with the recurrence $X_{n+1} = 2X_n + X_{n-1}$ and the identity $V_n^2 - 8U_n^2 = 4(-1)^n$. The Jacobi symbol is written (a/n) .

2. CHEBYSHEV BASES AND THE CHUA FRAMEWORK

2.1. The Chua identity. For an integer $a \neq 0, \pm 1$, set $\omega_a = a + \sqrt{a^2 - 1}$, so that $\omega_a \bar{\omega}_a = 1$ and $\omega_a^n + \omega_a^{-n} = 2T_n(a)$ where T_n is the n -th Chebyshev polynomial of the first kind — hence the name “Chebyshev base”.

Theorem 2.1 (Chua [5]; cf. Williams [8, Ch. 4]). *Let Q be an odd prime with $\gcd(a^2 - 1, Q) = 1$. Set $D = a^2 - 1$, $\varepsilon = (D/Q)$, $\delta = (2(a+1)/Q)$. Then*

$$\omega_a^{(Q-\varepsilon)/2} \equiv \delta \pmod{Q}$$

in $\mathbb{Z}[\sqrt{D}]/(Q)$.

Proof. Define $\gamma = a + 1 + \sqrt{D}$. Then $\gamma^2 = 2(a+1) \cdot \omega_a$. Compute γ^Q in $\mathbb{Z}[\sqrt{D}]/(Q)$ by Frobenius ($\gamma^Q = a + 1 + \varepsilon\sqrt{D}$) and via the square ($\gamma^Q = \delta \cdot \omega_a^{(Q-1)/2} \cdot \gamma$). Equating and dividing by γ (a unit) yields $\omega_a^{(Q-\varepsilon)/2} = \delta$. $\square$

2.2. Basic properties of Wagstaff numbers.

Lemma 2.2 (Order of 2). *For every prime $p \geq 5$ and every prime $r \mid W_p$: $\text{ord}_r(2) = 2p$ and $r \equiv 1 \pmod{2p}$.*

Proof. Since $3W_p = 2^p + 1$: $2^p \equiv -1 \pmod{r}$. Therefore $\text{ord}_r(2) \mid 2p$ but $\text{ord}_r(2) \nmid p$. Since p is prime, $\text{ord}_r(2) \in \{2, 2p\}$. If $\text{ord}_r(2) = 2$: $r \mid 3$, contradicting $\gcd(W_p, 3) = 1$ (since $p \geq 5$ is prime, $p \nmid 6 = \text{ord}_3(2)$, so $v_3(2^p + 1) = 1$ and $3 \nmid W_p$). So $\text{ord}_r(2) = 2p$, and $2p \mid r - 1$ by Fermat. $\square$

Lemma 2.3 (mod-8 classification). *For every odd prime $p \geq 3$:*

- (i) $W_p \equiv 3 \pmod{8}$;
- (ii) every prime factor of W_p satisfies $r \equiv 1$ or $3 \pmod{8}$;
- (iii) the total multiplicity of prime factors $\equiv 3 \pmod{8}$ is odd.

Proof. (i) Since p is odd, $3W_p = 2^p + 1 \equiv 1 \pmod{8}$, so $W_p \equiv 3 \pmod{8}$. (ii) By Lemma 2.2, $v_2(\text{ord}_r(2)) = v_2(2p) = 1$. For $r \equiv 7 \pmod{8}$: $(2/r) = 1$, so $\text{ord}_r(2) \mid (r-1)/2$ is odd, contradicting $v_2 = 1$. For $r \equiv 5 \pmod{8}$: $v_2(\text{ord}_r(2)) \leq v_2(r-1) = 2 \neq 1$. (iii) Write $W_p = \prod r_i^{e_i}$. By (ii), each $r_i \equiv 1$ or $3 \pmod{8}$. Modulo 8 the product is $\equiv 3^{\sum' e_i}$ where $\sum'$ runs over $r_i \equiv 3$. From (i): $\sum' e_i$ is odd. $\square$

2.3. The base $a = 3$.

Proposition 2.4 (Base $a = 3$ necessity). *For every Wagstaff prime W_p with $p \geq 5$,*

$$\omega_3^{(W_p+1)/2} \equiv -1 \pmod{W_p}, \quad \omega_3 = 3 + 2\sqrt{2}.$$

Proof. For $a = 3$: $D = a^2 - 1 = 8$ and $2(a+1) = 8$, so $\varepsilon = \delta = (8/W_p) = (2/W_p)^3 = (2/W_p)$. By Lemma 2.3 (i), $W_p \equiv 3 \pmod{8}$, so the second supplement to quadratic reciprocity gives $(2/W_p) = -1$; hence $\varepsilon = \delta = -1$. Theorem 2.1 with $Q = W_p$ yields $\omega_3^{(W_p+1)/2} \equiv -1 \pmod{W_p}$. $\square$

Remark 2.5 (Uniqueness). The base $a = 3$ is the unique integer $a \geq 2$ with $a^2 - 1$ a power of 2. Indeed, $(a-1)(a+1) = 2^k$ with $a-1, a+1$ two consecutive even numbers forces $\gcd(a-1, a+1) = 2$, so both are powers of 2 differing by 2 — hence $\{a-1, a+1\} = \{2, 4\}$ and $a = 3$. Consequently, $a = 3$ gives the *only* universal $N+1$ condition of the Chua form.

The identity that powers our method:

Lemma 2.6 (Fundamental identity). *For every $p \geq 5$:*

$$W_p + 1 = 4W_{p-2}.$$

Proof. $W_p + 1 = (2^p + 1)/3 + 1 = (2^p + 4)/3 = 4 \cdot (2^{p-2} + 1)/3 = 4W_{p-2}$. $\square$

3. PRIMALITY CRITERIA

3.1. Order lemma.

Lemma 3.1. *Let N be odd with $N \equiv 3 \pmod{8}$. Suppose $\omega_3^{N+1} \equiv 1$ and $\omega_3^{(N+1)/2} \equiv -1 \pmod{N}$. Then for every prime $r \mid N$: $v_2(\text{ord}_r(\omega_3)) = v_2(N + 1) = 2$.*

Proof. The congruences descend from mod N to mod r . Let $d = \text{ord}_r(\omega_3)$. From the first hypothesis: $d \mid N + 1$. From the second: $d \nmid (N + 1)/2$. Hence $v_2(d) = v_2(N + 1)$. Since $N \equiv 3 \pmod{8}$: $N + 1 \equiv 4 \pmod{8}$, so $v_2(N + 1) = 2$. $\square$

3.2. The $N + 1$ criterion.

Theorem 3.2 (General $N + 1$ criterion). *Let $p \geq 5$ be prime, $N = W_p$. Suppose Conditions (I) $\omega_3^{N+1} \equiv 1$ and (II) $\omega_3^{(N+1)/2} \equiv -1 \pmod{N}$ hold. Let S be a set of odd prime divisors of W_{p-2} with $\omega_3^{(N+1)/q} - 1$ a unit modulo N for each $q \in S$. Define $F = 4 \prod_{q \in S} q^{v_q(W_{p-2})}$. If $F > \sqrt{N}$, then N is prime.*

Proof. Suppose N is composite with smallest prime factor $r \leq \sqrt{N}$.

Step 1 (Order constraints). By Lemma 3.1, $d := \text{ord}_r(\omega_3)$ satisfies $d \mid N + 1$ and $v_2(d) = 2$ (since $N + 1 = 4W_{p-2}$ with W_{p-2} odd).

Step 2 (Full multiplicity: $F \mid d$). For each $q \in S$, the unit condition implies $\omega_3^{(N+1)/q} \not\equiv 1 \pmod{r}$. Since $d \mid N + 1$ but $d \nmid (N + 1)/q$: $v_q(d) = v_q(N + 1) = v_q(W_{p-2})$. Together with $v_2(d) = 2 = v_2(4)$: $F \mid d$.

Step 3 (Group-theoretic bound). Since $\text{Norm}(\omega_3) = 9 - 8 = 1$ and r is odd prime:

- If $(2/r) = -1$: $\mathbb{Z}[\sqrt{2}]/(r) \cong \mathbb{F}_{r,2}$ and ω_3 lies in the norm-1 subgroup μ_{r+1} , so $d \mid r + 1$.
- If $(2/r) = +1$: $\mathbb{Z}[\sqrt{2}]/(r) \cong \mathbb{F}_r \times \mathbb{F}_r$ and $d \mid r - 1$.

Case 1 ($d \mid r - 1$): $F \mid d \mid r - 1$, so $r \geq F + 1 > \sqrt{N} + 1$. Contradicts $r \leq \sqrt{N}$.

Case 2 ($d \mid r + 1$) — boundary analysis. From $F \mid d \mid r + 1$: $r \geq F - 1$. Since $N \equiv 3 \pmod{8}$, $\sqrt{N}$ is irrational. With $F > \sqrt{N}$ integer: $F \geq \lfloor \sqrt{N} \rfloor + 1$. Combined with $r \leq \lfloor \sqrt{N} \rfloor$: $\lfloor \sqrt{N} \rfloor \geq r \geq F - 1 \geq \lfloor \sqrt{N} \rfloor$, so $r = \lfloor \sqrt{N} \rfloor$ and $d = r + 1$.

Write $N = r \cdot s$ with $s \geq r$. From $(r + 1) \mid (rs + 1)$ and $r \equiv -1 \pmod{r + 1}$: $s \equiv 1 \pmod{r + 1}$. The smallest candidate is $s = r + 2$; the next, $s = 2r + 3$, gives $N = r(2r + 3) = 2r^2 + 3r > r^2 + 2r + 1 = (r + 1)^2$, contradicting $r = \lfloor \sqrt{N} \rfloor$. So $N = r(r + 2)$, $N + 1 = (r + 1)^2$. Since $N + 1 = 4W_{p-2}$: $W_{p-2} = ((r + 1)/2)^2$. But $W_{p-2} \equiv 3 \pmod{8}$ (Lemma 2.3), and a perfect square is $\equiv 0, 1, 4 \pmod{8}$. Contradiction. $\square$

3.3. Coprimality and the W_{p-2} -prime case.

Corollary 3.3 (Coprimality). *Every prime divisor of W_{p-2} is coprime to W_p .*

Proof. $3W_p - 12W_{p-2} = (2^p + 1) - 4(2^{p-2} + 1) = -3$, so $W_p - 4W_{p-2} = -1$. $\square$

Corollary 3.4. *If W_{p-2} is prime, then W_p is prime iff $\omega_3^{(W_p+1)/2} \equiv -1 \pmod{W_p}$.*

Proof. Necessity: Proposition 2.4. *Sufficiency:* let $r \mid N$. By Lemma 3.1, $d := \text{ord}_r(\omega_3)$ has $v_2(d) = 2$ and $d \mid N + 1 = 4W_{p-2}$. The divisors of $4W_{p-2}$ with $v_2 = 2$ are 4 and $4W_{p-2}$. We exclude $d = 4$: $\omega_3^4 = 577 + 408\sqrt{2}$, and $r \mid \gcd(576, 408) = 24$ contradicts $\gcd(W_p, 24) = 1$. So $d = N + 1$. Since $d \mid r \pm 1$: $r \geq N$, forcing $r = N$. $\square$

Remark 3.5. Corollary 3.4 applies directly to all small- p Wagstaff primes where W_{p-2} is known prime: $p = 5$ (with $W_3 = 3$ prime), $p = 7$ ($W_5 = 11$), $p = 13$ ($W_{11} = 683$), $p = 19$ ($W_{17} = 43,691$), and ascends through the 29 Wagstaff primes of §A as far as $p - 2$ also lies in that list. The criterion in Corollary 3.4 is a deterministic primality test for W_p in every such case; no factorisation beyond confirming W_{p-2} prime is required.

3.4. The BLS $N - 1$ theorem. For the three target exponents, W_{p-2} is not known to have enough factored content to satisfy $F_+ > \sqrt{W_p}$ in Theorem 3.2; the $N + 1$ factored portion is $F_+ = 4$ at best, because full factorisations of W_{p-2} are unavailable at these sizes. The proofs in Section 4 therefore work on the $N - 1$ side using the classical BLS theorem, which we recall:

Theorem 3.6 (BLS $N - 1$, [3, Theorem 5; strengthened form, Theorem 11]). *Let $N > 1$ be an odd integer. Write $N - 1 = FR$ with $F > 1$, $\gcd(F, R) = 1$, and F completely factored into known primes. Suppose for each prime $q \mid F$ there exists an integer a_q with*

$$a_q^{N-1} \equiv 1 \pmod{N} \quad \text{and} \quad \gcd(a_q^{(N-1)/q} - 1, N) = 1.$$

If $F > \sqrt{N}$, then N is prime. More generally, if $F > N^{1/3}$, write $N = rF + s + 1$ with $0 \leq s < F$ and set $\Delta = s^2 - 8r$; if Δ is not a perfect square, then N is prime.

The $F > N^{1/3}$ threshold with the discriminant check ([3, Theorem 11]) is BLS's central improvement over the classical $F > N^{1/2}$: instead of needing half the bit-size of N to be factored, we need only a third.

4. THE THREE PRIMALITY PROOFS

4.1. Framework. The three proofs in §§4.2–4.4 invoke Theorem 3.6 (BLS $N - 1$) with specific factorisations of $N - 1 = 2(2^{p-1} - 1)/3$ drawn from the cyclotomic decomposition

$$2^{p-1} - 1 = \prod_{d \mid p-1} \Phi_d(2),$$

harvested partly from the Cunningham project tables [4] and partly from direct computation. Each factor of each $\Phi_d(2)$ contributes to the BLS

factored portion $F = F_-$ once its primality is certified by APR-CL [1, 6]; Fermat-type witnesses $a \in \{2, 3, 5, \dots\}$ are then validated for each prime $q \mid F$. The exact software toolchain and version numbers used to produce the certificates are documented in Section 5. The proofs are unconditional and do not rely on the Chebyshev sufficiency conjecture.

At each p , we also independently verify the Chua $N + 1$ condition $\omega_3^{(W_p+1)/2} \equiv -1 \pmod{W_p}$ (Proposition 2.4) as a sanity check on the modular-arithmetic pipeline. This check is consistent with the BLS proof (both confirm W_p prime) but does not itself enter the BLS argument.

APR-CL certification is applied uniformly to every prime q of F , regardless of size — the smallest single-digit primes receive the same unconditional APR-CL treatment as the largest multi-hundred-digit factors. The “exact margin” column reports $(2F^3)_{\text{bits}} - N_{\text{bits}}$, the integer bit-margin by which the BLS hypothesis $2F^3 > N$ is satisfied.

p	W_p digits	div. of $p-1$	primes in F	exact margin (bits)	wall time
2617	788	16	22	47	< 10 s (Ryzen 5 3600)
10501	3161	48	103	3262	98 s (1 core, 3.6 GHz)
12391	3730	32	61	2861	82 s (1 core, 3.6 GHz)

For each proof, Condition (II) is independently verified ($\omega_3^{(N+1)/2} \equiv -1 \pmod{N}$) as a sanity check; full data in `bls_certificate_w{2617,10501,12391}.json` (Section 5).

4.2. W_{2617} .

Theorem 4.1. *The Wagstaff number W_{2617} is prime.*

Proof. Set $N = W_{2617}$, a 788-digit number. We apply Theorem 3.6 to the $N - 1$ side; the factored portion F will be shown to exceed $N^{1/3}$.

The exponent satisfies $p - 1 = 2616 = 2^3 \cdot 3 \cdot 109$, with 16 divisors. The cyclotomic decomposition

$$2^{p-1} - 1 = \prod_{d \mid p-1} \Phi_d(2)$$

gives 16 factors, of which 12 (those with $d \leq 654$) are fully factored by direct computation. Their cumulative product contributes 22 prime factors to F (a factor of $(N - 1)/2$), and $2F^3 > N$ by a 47-bit margin.

For each prime $q \mid F$ (the 22 primes constituting the factored portion), the witness $a_q \in \{2, 3\}$ is tested: $a_q^{N-1} \equiv 1 \pmod{N}$ and $\gcd(a_q^{(N-1)/q} - 1, N) = 1$. Every witness succeeds (full data in `bls_certificate_w2617.json`). Writing $N = rF + s + 1$ with $0 \leq s < F$ and $\Delta = s^2 - 8r$, the discriminant Δ is not a perfect square.

By Theorem 3.6, $N = W_{2617}$ is prime. $\square$

Remark 4.2. As a sanity check, the Chua $N + 1$ condition $\omega_3^{(N+1)/2} \equiv -1 \pmod{N}$ is verified independently: the modular exponentiation in $\mathbb{Z}[\sqrt{2}]/(N)$ takes under 1 second with gmpy2.

4.3. W_{10501} .

Theorem 4.3. *The Wagstaff number W_{10501} is prime.*

Proof. Set $N = W_{10501}$, a 3161-digit number. The exponent $p - 1 = 10500 = 2^2 \cdot 3 \cdot 5^3 \cdot 7$ has 48 divisors. Of the 48 cyclotomic factors $\Phi_d(2)$ of $2^{p-1} - 1$, 43 are fully factored — 38 by direct computation (using SymPy `factorint` with Pollard ρ , $p - 1$, and ECM) and 5 retrieved from the Cunningham project tables [4]. This contributes 103 primes to the BLS factored portion F , each certified prime by APR-CL [1, 6].

The factored portion satisfies $2F^3 > N$ by a margin of 3262 bits, substantially exceeding the BLS threshold. Fermat-type witnesses $a_q \in \{2, 3, 5\}$ are validated for each of the 103 primes $q \mid F$; every witness succeeds. The discriminant $\Delta = s^2 - 8r$ (with $N = rF + s + 1$, $0 \leq s < F$) is computed and verified non-square. Total wall time: 98s on one core of a 3.6 GHz machine; full data in `bls_certificate_w10501.json`.

By Theorem 3.6, $N = W_{10501}$ is prime. $\square$

4.4. W_{12391} .

Theorem 4.4. *The Wagstaff number W_{12391} is prime.*

Proof. Set $N = W_{12391}$, a 3730-digit number. The exponent $p - 1 = 12390 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 59$ has 32 divisors. Of the 32 cyclotomic factors $\Phi_d(2)$, 27 are fully factored: 19 by direct computation, 7 retrieved from the Cunningham tables [4] and public factor databases (every such factor re-certified by APR-CL [1, 6] before use), and 1 itself prime. This contributes 61 primes in F_- , the largest being a 371-digit prime cofactor of $\Phi_{2065}(2)$. Each of the 61 primes receives independent APR-CL certification.

The factored portion F satisfies $2F^3 > N$ by a 2861-bit margin. Fermat-type witnesses $a_q \in \{2, 3\}$ are validated for each of the 61 primes; every witness succeeds. The discriminant $\Delta = s^2 - 8r$ is non-square. Total wall time: 82s on one core of a 3.6 GHz machine (less than W_{10501} despite the larger N , because $p - 1$ has fewer but larger prime factors: less divisor enumeration, although each factorisation task is heavier). Full data in `bls_certificate_w12391.json`.

By Theorem 3.6, $N = W_{12391}$ is prime. $\square$

4.5. Limits of the method.

Remark 4.5 (BLS ceiling). The BLS $N - 1$ approach requires $p - 1$ to be sufficiently smooth that enough cyclotomic factors $\Phi_d(2)$ with $d \mid p - 1$ have known factorisations. Among the 36 known Wagstaff primes and probable primes with $p > 2617$, the exponents $p = 10501$ and $p = 12391$ are the only ones meeting this threshold in the current Cunningham tables. The next candidate by smoothness, $p = 14479$ (with $p - 1 = 2 \cdot 3 \cdot 19 \cdot 127$), falls several thousand bits short; the main missing piece is a partial factorisation of $\Phi_{14478}(2)$. Extensions of the Cunningham tables [4] — or direct ECM on the divisors $\Phi_d(2)$ for $d \mid p - 1$ with the target exponents p — would push the feasibility frontier. The cofactor data recorded in the JSON certificates (Section 5) identifies exactly which $\Phi_d(2)$ remain only partially factored for $p = 10501$ and $p = 12391$; these are natural targets for further ECM work.

5. COMPUTATIONAL DETAILS

5.1. Software toolchain. All computational components in this paper are reproducible from the scripts in the supplementary repository using free, open-source software.

- *Integer factorisation* (cyclotomic factors $\Phi_d(2)$): SymPy ≥ 1.12 `factorint` (combining trial division, Pollard ρ , Pollard $p-1$, and ECM internally), augmented for cyclotomic factors by trial division along the arithmetic progression $r \equiv 1 \pmod{d}$ implied by primitive-divisor theory.
- *Pre-known large factorisations.* The Cunningham project tables [4] for cyclotomic factors $\Phi_d(2)$ with $d \leq 1200$, and FactorDB as a discovery aid for two larger terms ($\Phi_{2065}(2)$ and $\Phi_{2478}(2)$ in Theorem 4.4). Every retrieved factor is independently re-verified for primality by APR-CL before use; FactorDB plays no evidentiary role.
- *Primality of cofactors* (all factors used in BLS certificates): the APR-CL implementation in PARI/GP ≥ 2.15 (`isprime(n, 2)`), which forces the APR-CL (Adleman–Pomerance–Rumely / Cohen–Lenstra) path and is unconditionally deterministic [1, 6]. The largest cofactor is a 371-digit factor of $\Phi_{2065}(2)$ (Theorem 4.4); the smallest are single-digit primes, but all receive APR-CL certification for uniformity.
- *Fast modular arithmetic* (the modular exponentiations $a^{N-1} \bmod N$ and $\omega_3^{(N+1)/2} \bmod N$ for $N = W_{10501}, W_{12391}$): Python’s built-in big-int `pow`, with GMP-backed acceleration via `gmpy2` ≥ 2.1 when installed.

The driver scripts `bls_n_minus_1_w{2617,10501,12391}.py` in the supplementary repository emit JSON certificates (`bls_certificate_w*.json`) recording every cyclotomic factor, prime decomposition, BLS witness, and primality method used, along with full execution logs. A master verifier `verify_bls.py` re-reads each certificate from disk, treats every claim in it as untrusted input, rebuilds F from $v_q(N-1)$, re-runs APR-CL on every prime, re-checks the witnesses and the discriminant, and re-runs Condition (II); any disagreement exits non-zero.

5.2. Supplementary material. All data and scripts are archived on Zenodo at <https://doi.org/10.5281/zenodo.19643792> (concept DOI; always resolves to the latest version), with a development mirror on GitHub at <https://github.com/dolonet/wagstaff-bls-primality>.

The archive contains, for each $p \in \{2617, 10501, 12391\}$:

- (1) the BLS primality certificate (JSON), recording every cyclotomic factor $\Phi_d(2)$, its prime decomposition, the BLS witness a for each prime $q \mid F$, and the APR-CL timings for each cofactor;
- (2) the driver script `bls_n_minus_1_w<p>.py` reproducing the certificate end-to-end;
- (3) the full APR-CL primality log for each cofactor, reproducible via `gp -q isprime.gp`;
- (4) the independent Condition (II) verification log.

The archive is deterministic and version-pinned; running the driver scripts on a clean machine reproduces every numeric result in this paper. The master verifier `verify_bls.py` reads each certificate and re-derives F , the margin, witnesses, discriminant, and Condition (II) from scratch.

6. RELATION TO THE CHEBYSHEV SUFFICIENCY CONJECTURE

The content of Sections 1–5 is independent of the Chebyshev sufficiency conjecture — the question of whether the single congruence $\omega_3^{(W_p+1)/2} \equiv -1 \pmod{W_p}$ alone suffices to decide primality of W_p . That conjecture is necessary (Proposition 2.4) and conjecturally sufficient; its current structural status — including a reduction to the Pell Primitive Pair Conjecture (PPPC) — is the subject of the companion paper [9]. A further companion paper [10] gives a universal unconditional Frobenius primality test in $\mathbb{Z}[\sqrt{3}]$ for the broader family $(b^p \pm 1)/(b \pm 1)$; the ring and the base there are different from the ones used in this paper ($\mathbb{Z}[\sqrt{2}]$ and $\omega_3 = 3 + 2\sqrt{2} = (1 + \sqrt{2})^2$ here vs. $\mathbb{Z}[\sqrt{3}]$ and $\omega_3 = (2 + \sqrt{3})^2 = 7 + 4\sqrt{3}$ there).

APPENDIX A. REFERENCE DATA

A.1. The 36 known Wagstaff primes and probable primes. Proved prime ($p \leq 42737$, by ECPP [2] or by this paper), excluding the trivial case $p = 3$ for which $W_3 = 3$:

$$p \in \{5, 7, 11, 13, 17, 19, 23, 31, 43, 61, 79, 101, 127, 167, 191, 199, 313, 347, 701, 1709,$$

$$2617, 3539, 5807, 10501, 10691, 11279, 12391, 14479, 42737\}$$

— 29 exponents. Probable prime ($p > 42737$, not yet proved but consistent with every applied pseudoprime test):

$$p \in \{83339, 95369, 117239, 127031, 138937, 141079, 267017\}$$

— 7 exponents. The “36 known” count in the header includes both groups; adding $p = 3$ would make 37.

A.2. Cyclotomic decomposition of $W_p \pm 1$. For odd prime p and $N = W_p$:

$$N - 1 = \frac{2^p - 2}{3} = \frac{2(2^{p-1} - 1)}{3}, \quad N + 1 = 4W_{p-2}.$$

On the $N - 1$ side the cyclotomic decomposition

$$2^{p-1} - 1 = \prod_{d|p-1} \Phi_d(2)$$

harvests factored portions from divisors d of $p - 1$; each $\Phi_d(2)$ is a positive integer whose prime factors lie in an arithmetic progression $r \equiv 1 \pmod{d}$. This is the source of the BLS factored portions in §§4.2–4.4.

On the $N + 1$ side the identity $W_p + 1 = 4W_{p-2}$ (Lemma 2.6) gives a single cofactor W_{p-2} whose factorisation is harder in general but tractable for small p ; Theorem 3.2 is phrased in terms of this decomposition.

REFERENCES

- [1] L. M. Adleman, C. Pomerance, R. S. Rumely, *On distinguishing prime numbers from composite numbers*, Ann. of Math. (2) **117** (1983), 173–206.
- [2] A. O. L. Atkin, F. Morain, *Elliptic curves and primality proving*, Math. Comp. **61** (1993), no. 203, 29–68.
- [3] J. Brillhart, D. H. Lehmer, J. L. Selfridge, *New primality criteria and factorizations of $2^m \pm 1$* , Math. Comp. **29** (1975), no. 130, 620–647.
- [4] J. Brillhart, D. H. Lehmer, J. L. Selfridge, B. Tuckerman, S. S. Wagstaff, Jr., *Factorizations of $b^n \pm 1$, $b = 2, 3, 5, 6, 7, 10, 11, 12$ Up to High Powers*, 3rd ed., Contemporary Mathematics, vol. 22, American Mathematical Society, 2002. Online updates at <https://homes.cerias.purdue.edu/~ssw/cun/>.
- [5] K. S. Chua, *Chebyshev polynomials and higher order Lucas–Lehmer algorithm*, preprint arXiv:2010.02677, 2020.
- [6] H. Cohen, A. K. Lenstra, *Implementation of a new primality test*, Math. Comp. **48** (1987), no. 177, 103–121.
- [7] D. H. Lehmer, *On Lucas’s test for the primality of Mersenne’s numbers*, J. London Math. Soc. **10** (1935), no. 3, 162–165.
- [8] H. C. Williams, *Édouard Lucas and Primality Testing*, CMS Series of Monographs and Advanced Texts **22**, Wiley-Interscience, 1998.
- [9] A. Dolotov, *Reduction of the Wagstaff Chebyshev sufficiency conjecture to the Pell Primitive Pair Conjecture*, companion paper (in preparation).
- [10] A. Dolotov, *A deterministic Frobenius primality test in $\mathbb{Z}[\sqrt{3}]$ for the family $(b^p \pm 1)/(b \pm 1)$* , companion paper (in preparation).

Email address: science@dolotov.com

URL: <https://orcid.org/0009-0003-9481-5808>